

JHOAN FRANCO

Software Engineer

Dual engineering degree in progress

Backend | Data engineering | Distributed systems

Availability: 4–5 month internship

+33 7 52 91 66 63
jhoan.franco-ruiz@imt-atlantique.net
GitHub LinkedIn
Portfolio
Brest, France

Profile

Software developer with experience in backend and full stack development, currently pursuing a dual engineering degree at IMT Atlantique and Universidad Nacional de Colombia. Skilled in REST API design and in integrating cloud and artificial intelligence services, with a strong interest in software architecture, distributed systems and application reliability.

Education

•IMT Atlantique *In progress (expected 2028)*

Master of Engineering (Diplôme d'ingénieur) – Specialization in distributed systems Brest, France

•Universidad Nacional de Colombia *In progress (expected 2028)*

Systems and Computer Engineering (5-year degree) Bogotá, Colombia

Professional Experience

•Software Engineering Intern (Full Stack) *Mar. – Jul. 2026*

Yunex Traffic Bogotá, Colombia

- Developed the backend (Java, Spring Boot) and frontend (React) of a maintenance module integrated into ORION, used to track field operations: events, groups and inventory of intelligent transportation equipment.
- Improved inventory traceability through REST APIs and a hexagonal architecture, building a scalable solution designed to be sold as a software product beyond internal use.
- Managed deliverables according to the company's software engineering standards.

•Software Engineer *Aug. 2025 – Feb. 2026*

Nodi (start-up) Bogotá, Colombia

- Migrated all data from Supabase to PostgreSQL on GCP to strengthen the cloud architecture.
- Designed a Google Cloud Storage bucket structure to organize résumés, PDFs and images.
- Developed backend services in Python with FastAPI, integrating Google's AI APIs.

•Programming Teaching Assistant *Feb. – May and Jul. – Sep. 2025*

Universidad Nacional de Colombia Bogotá, Colombia

- Supported 120 students in C++ programming (2 hours a day, 2 days a week) as part of a team of 3 teaching assistants: graded exercises and reinforced core concepts.

Projects

•CDC Architecture for Data Integration *Data engineering, ETL/ELT, data pipelines* *Sep. 2026 – present*

- Designing a real-time CDC architecture POC (Debezium, Apache Pulsar, Apache Flink) for incremental ingestion from multiple sources into a data warehouse, for **Clever Cloud**.
- Experimental evaluation of the pipeline: consistency and reliability testing.

•Academic Management System (SGA) *Microservices, DevOps, social impact* *May – Jul. 2025*

- Built, in a team of 7, a web **MVP** for Colegio Josué Manrique, replacing an Access/Excel process: pre-enrollment with PDF generation, role-based authentication, grade and attendance tracking (Medium article).
- SOFEA microservices architecture (FastAPI, React, MongoDB, MySQL) with CI/CD on GitHub Actions, Docker, and Prometheus/Grafana monitoring.

Technical Skills

Languages: Java, Python, JavaScript, C++, SQL, Go, Elixir, R.

Backend and frontend development: FastAPI, Spring Boot, Django, React, REST APIs, JWT, HTML, CSS, Bootstrap.

Software architecture and distributed systems: microservices, hexagonal architecture, SOFEA, monolithic and decoupled architectures.

Data engineering and databases: CDC, Debezium, Apache Pulsar, Apache Flink, Apache Spark, ETL/ELT, data pipelines, PostgreSQL, MySQL, MongoDB, Redis.

Cloud and DevOps: Google Cloud Platform (Cloud Storage, Vertex AI, Cloud Run, Compute Engine, Cloud SQL), Firebase, Docker, Kubernetes (Minikube, k0s), GitHub Actions, Prometheus, Grafana, Linux/Bash.

Testing, tools and methods: JUnit, Mockito, Pytest, Selenium, SonarQube, Git, GitHub, GitLab, Jira, Confluence, Scrum.

Interests: new technologies, start-up ecosystem and entrepreneurship, football, language learning and discovering new cultures.

Languages

Spanish: native **French:** B2 (DELF certified) **English:** B2 (upper intermediate)